# 使用统计信息通过 SMB 监控 Hyper-V 和 SQL Server 活动

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# 使用统计信息通过 SMB 监控 Hyper-V 和 SQL Server 活动

## 确定可用的统计信息对象和计数器

在获取有关 CIFS ， SMB ，审核和 BranchCache 哈希统计信息以及监控性能的信息之前，您必须了解哪些对象和计数器可用于获取数据。

步骤

1. 将权限级别设置为高级：

   ` * 设置 -privilege advanced*`

2. 执行以下操作之一：

| 要确定的内容 | 输入 ... |
|---|---|
| 哪些对象可用 | ` * 统计信息目录对象 show*` |
| 可用的特定对象 | ` * 统计信息目录对象 show object *object_name**` |
| 哪些计数器可用 | ` * 统计信息目录计数器 show object *object_name**` |

   有关哪些对象和计数器可用的详细信息，请参见手册页。

3. 返回到管理权限级别：

   ` * 设置 -privilege admin*`

以下命令显示与集群中的 CIFS 和 SMB 访问相关的选定统计信息对象的说明，如高级权限级别所示：

```
cluster1::> set -privilege advanced

Warning: These advanced commands are potentially dangerous; use them only
when directed to do so by support personnel.
Do you want to continue? {y|n}: y

cluster1::*> statistics catalog object show -object audit
    audit_ng                    CM object for exporting audit_ng
performance counters

cluster1::*> statistics catalog object show -object cifs
    cifs                        The CIFS object reports activity of the
                                Common Internet File System protocol
                                ...

cluster1::*> statistics catalog object show -object nblade_cifs
    nblade_cifs                 The Common Internet File System (CIFS)
                                protocol is an implementation of the
Server
                                ...

cluster1::*> statistics catalog object show -object smb1
    smb1                        These counters report activity from the
SMB
                                revision of the protocol. For information
                                ...

cluster1::*> statistics catalog object show -object smb2
    smb2                        These counters report activity from the
                                SMB2/SMB3 revision of the protocol. For
                                ...

cluster1::*> statistics catalog object show -object hashd
    hashd                       The hashd object provides counters to
measure
                                the performance of the BranchCache hash
daemon.
cluster1::*> set -privilege admin
```

以下命令显示有关在高级权限级别看到的 `CIFS` 对象的某些计数器的信息：

> 此示例不会显示 `cifs` 对象的所有可用计数器；输出将被截断。

```
cluster1::> set -privilege advanced

Warning: These advanced commands are potentially dangerous; use them only
when directed to do so by support personnel.
Do you want to continue? {y|n}: y

cluster1::*> statistics catalog counter show -object cifs

Object: cifs
    Counter                     Description
    ---------------------------
----------------------------------------------
    active_searches             Number of active searches over SMB and
SMB2
    auth_reject_too_many        Authentication refused after too many
                                requests were made in rapid succession
    avg_directory_depth         Average number of directories crossed by
SMB
                                and SMB2 path-based commands
    ...                         ...

cluster2::> statistics start -object client -sample-id
Object: client
    Counter                                                     Value
    -------------------------------- --------------------------------
    cifs_ops                                                        0
    cifs_read_ops                                                   0
    cifs_read_recv_ops                                              0
    cifs_read_recv_size                                            0B
    cifs_read_size                                                 0B
    cifs_write_ops                                                  0
    cifs_write_recv_ops                                             0
    cifs_write_recv_size                                           0B
    cifs_write_size                                                0B
    instance_name                             vserver_1:10.72.205.179
    instance_uuid                                     2:10.72.205.179
    local_ops                                                       0
    mount_ops                                                       0


[...]
```

# 显示 SMB 统计信息

## 您可以显示各种 SMB 统计信息来监控性能和诊断问题。

步骤

1. 使用 `satictics start` 和可选的 `satictics stop` 命令收集数据样本。

   有关这些命令的详细信息，请参见 "《系统管理参考》"。

2. 执行以下操作之一：

| 要显示统计信息的对象 | 输入以下命令 ... |
|---|---|
| SMB 的所有版本 | ` * statistics show -object cifs*` |
| SMB 1.0 | ` * statistics show -object smb1*` |
| SMB 2.x 和 SMB 3.0 | ` * statistics show -object smb2*` |
| 节点的 CIFS 子系统 | ` * statistics show -object nblade_cifs*` |

有关详细信息，请参见手册页。

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